

PREDICTION INTERVAL FOR A MIXTURE OF WEIBULL FAILURE-TIME DISTRIBUTIONS – A BAYESIAN APPROACH

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Summary: The predictive distribution and predictive interval of a future observation from a mixture of two Weibull distributions are studied with a Monte-Carlo Simulation, using a Bayesian approach.

1. INTRODUCTION

In life testing problems, one often encounters situations where the underlying distribution is not homogeneous, but may consist of several subpopulations mixed in unknown proportions. Consider a case where a warehouse of electrical components may be comprised of two subpopulations, each with a different life expectancy. For example, a manufacturer may produce a brand name line as well as a generic product. Each item of the population may be attributed to the appropriate subpopulation with the subpopulations being mixed in unknown proportions α and $1 - \alpha = \beta$.

The component lifetimes we consider here have a mixture distribution

$$F(t) = \alpha F_1(t) + \beta F_2(t) \text{ where}$$
$$F_i(t) = 1 - \exp\left(-\frac{t^{p_i}}{\theta_i}\right) \quad i = 1, 2$$

with the corresponding density function

$$(1.1) \quad f(t|\theta_1, \theta_2, p_1, p_2, \alpha) = \frac{\alpha p_1}{\theta_1} t^{p_1-1} \exp\left(-\frac{t^{p_1}}{\theta_1}\right) \\ + \beta \frac{p_2}{\theta_2} t^{p_2-1} \exp\left(-\frac{t^{p_2}}{\theta_2}\right) \\ t, p_1, p_2, \theta_1, \theta_2 > 0, \quad \beta = 1 - \alpha, \quad \alpha < 1.$$

A considerable amount of research has been devoted to the study of mixture distributions. Sinha (1983) provides an extensive list.

2. PREDICTION INTERVAL: COMPLETE SAMPLES

Let $\underline{x}$ be a random sample from a distribution $F(x|\theta)$ characterized by a density function $f(x|\theta)$ where θ may be real or vector valued parameter. Let $\underline{y}$ be a second independent random sample of future observations from the same distribution. The Bayesian predictive distribution of a future observation Y is defined as the posterior expectation of $f(y|\theta)$ and is given by

$$(2.1) \quad h(y|\underline{x}) \propto \int_{\Omega} f(y|\theta) \Pi(\theta|\underline{x}) d\theta$$

where $\Pi(\theta|\underline{x})$ is the posterior distribution of θ and Ω is the range space of θ .

Predictive distributions have been studied by Aitchison (1964), Chhikara and Guttman (1982), Lawless (1977) and Sinha (1985), among others. For certain applications in reliability and quality control (Englehardt & Bain (1979), Lawless (1973) and McCool (1975, 1979) for example), it is desirable to obtain the confidence interval of a future random observation drawn from the underlying distribution. Such an interval is known as a prediction interval.

Mendenhall and Hader (1958) observed failure times of transmitter-receivers until a specified time T had elapsed. When brought in for maintenance the items were further checked and classified into two groups — unconfirmed failures and confirmed failures. Thus, once a failure occurred, each failed item may be attributed to the appropriate subpopulation.

In this paper, we assume n units from the Mendenhall and Hader (1958) mixture model (1.1) are subjected to some life testing experiment and let $\underline{x} = (x_1, x_2, \dots, x_n)$ be the failure times of these units. We further assume that n components of the same kind are to be put into future use and let $(y_1, y_2, \dots, y_n)$ be the future failure times of these components. We will study the predictive distribution and the corresponding prediction interval of the future failure time Y on the basis of the previously observed life test data $\underline{x}$ from the mixture density (1.1) using a Bayesian approach.

Let t_{ij} denote the failure time of the j th unit belonging to the i th subpopulation, $j = 1, 2, \dots, n_i, i = 1, 2$ and $\underline{t} = \{t_{ij} | j = 1, 2, \dots, n_i, i = 1, 2\}$ so that $n = n_1 + n_2$ observations are drawn from the mixture distribution (1.1). The likelihood

$$(2.2) \quad L \propto \frac{\alpha^{n_1}(1-\alpha)^{n_2} p_1^{n_1} p_2^{n_2}}{\theta_1^{n_1} \theta_2^{n_2}} \lambda_1^{p_1-1} \lambda_2^{p_2-1} \exp \left(-\frac{\sum_{j=1}^{n_1} t_{1j}^{p_1}}{\theta_1} \right) \\ \cdot \exp \left(-\frac{\sum_{j=1}^{n_2} t_{2j}^{p_2}}{\theta_2} \right)$$

where $\lambda_i = \prod_{j=1}^{n_i} (t_{ij})$.

We assume *a-priori* $p_i \sim U(0, \infty)$ and $\alpha \sim U(0, 1)$. Suppose little is known about θ_i so that Jeffreys (1961) 'vague prior' $p(\theta_1, \theta_2) \propto \frac{1}{\theta_1 \theta_2}$ would be appropriate. The joint prior distribution of the parameters may now be written as:

$$(2.3) \quad g(\theta_1, \theta_2, p_1, p_2, \alpha) \propto \frac{1}{\theta_1 \theta_2}.$$

From (2.2) and (2.3) we obtain the joint posterior distribution.

$$(2.4) \quad \prod(\theta_1, \theta_2, p_1, p_2, \alpha | \underline{t}) \propto \frac{\alpha^{n_1}(1-\alpha)^{n_2} p_1^{n_1} p_2^{n_2}}{\theta_1^{n_1+1} \theta_2^{n_2+1}} \lambda_1^{p_1-1} \lambda_2^{p_2-1} \\ \cdot \exp \left[-\frac{\sum_{j=1}^{n_1} t_{1j}^{p_1}}{\theta_1} - \frac{\sum_{j=1}^{n_2} t_{2j}^{p_2}}{\theta_2} \right].$$

Substituting (2.4) in (2.1), after some algebra, the predictive density of a future observation y is given by

(2.5)

$$h(y|\underline{t}) = C \left[n_1(n_1 + 1) \int_0^\infty \frac{\lambda_1^{p_1-1} p_1^{n_1+1} y^{p_1-1} dp_1}{\left\{ \sum_{j=1}^{n_1} t_{1j}^{p_1} + y^{p_1} \right\}^{n_1+1}} \int_0^\infty \frac{\lambda_2^{p_2-1} p_2^{n_2+1} y^{p_2-1} dp_2}{\left\{ \sum_{j=1}^{n_2} t_{2j}^{p_2} + y^{p_2} \right\}^{n_2+1}} + \right. \\ \left. n_2(n_2 + 1) \int_0^\infty \frac{\lambda_1^{p_1-1} p_1^{n_1} dp_1}{\left\{ \sum_{j=1}^{n_1} t_{1j}^{p_1} \right\}^{n_1}} \int_0^\infty \frac{\lambda_2^{p_2-1} p_2^{n_2+1} y^{p_2-1} dp_2}{\left\{ \sum_{j=1}^{n_2} t_{2j}^{p_2} + y^{p_2} \right\}^{n_2+1}} \right] \\ 0 < y < \infty$$

where

$$(2.6) \quad C^{-1} = (n+2) \left[\int_0^\infty \frac{\lambda_1^{p_1-1} p_1^{n_1} dp_1}{\left\{ \sum_{j=1}^{n_1} t_{1j}^{p_1} \right\}^{n_1}} \int_0^\infty \frac{\lambda_2^{p_2-1} p_2^{n_2} dp_2}{\left\{ \sum_{j=1}^{n_2} t_{2j}^{p_2} \right\}^{n_2}} \right]$$

Since the integrals (2.5) and (2.6) cannot be expressed in closed forms we resort to numerical integration techniques to compute the equal-tail- $(1-\alpha)$ -prediction interval $[C_1, C_2]$. C_1 and C_2 are solutions of

$$(2.7) \quad \frac{\alpha}{2} = \int_0^{C_1} h(y|\underline{t}) dy = \int_{C_2}^\infty h(y|\underline{t}) dy.$$

3. PREDICTION INTERVAL: CENSORED SAMPLE

Now suppose the test is terminated after a pre-assigned time T hours. Suppose r units have failed during the interval $(0, T)$, r_1 units from the first subpopulation and r_2 units from the second subpopulation so that $r = r_1 + r_2$.

Let $\underline{t} = (t_{11}, t_{12}, \dots, t_{1r_1}; t_{21}, t_{22}, \dots, t_{2r_2})$, $t_{ij} \leq T, j = 1, 2, \dots, r_i, i = 1, 2$. The likelihood function

$$(3.1) \propto \frac{\alpha^{r_1} (1-\alpha)^{r_2} p_1^{r_1} p_2^{r_2} \lambda_1^{p_1-1} \lambda_2^{p_2-1}}{\theta_1^{r_1} \theta_2^{r_2}} \exp \left\{ -\frac{\sum_{j=1}^{r_1} t_{1j}^{p_1}}{\theta_1} - \frac{\sum_{j=1}^{r_2} t_{2j}^{p_2}}{\theta_2} \right\} \\ \left\{ \alpha \exp \left(-\frac{T^{p_1}}{\theta_1} \right) + (1-\alpha) \exp \left(-\frac{T^{p_2}}{\theta_2} \right) \right\}^{(n-r)}$$

where $\lambda_i = \prod_{j=1}^{r_i} t_{ij}$.

The joint posterior distribution is given by:

$$(3.2) \quad \prod (\theta_1, \theta_2, p_1, p_2, \alpha | \underline{t}) \propto p_1^{r_1} p_2^{r_2} \lambda_1^{p_1-1} \lambda_2^{p_2-1} \\ \sum_{k=0}^{n-r} \binom{n-r}{k} \alpha^{n-k-r_2} (1-\alpha)^{r_2+k} \\ \left[\exp \left\{ -\frac{\sum_{j=1}^{r_1} t_{1j}^{p_1} + (n-r-k)T^{p_1}}{\theta_1} \right\} / \theta_1^{r_1+1} \right] \\ \exp \left\{ -\frac{\sum_{j=1}^{r_2} t_{2j}^{p_2} + kT^{p_2}}{\theta_2} \right\} / \theta_2^{r_2+1} \Bigg].$$

Substituting (3.2) in (2.1) and integrating out $(\theta_1, \theta_2, \alpha)$ the predictive distribution of a future observation Y is given by:

$$(3.3) \quad h(\underline{y} | \underline{t}) = C \sum_{k=0}^{n-r} \binom{n-r}{k} B(n-k-r_2+1, r_2+k+1) \\ \left[r_1(n-k-r_2+1) \int_0^\infty \frac{p_1^{r_1+1} \lambda_1^{p_1-1} y^{p_1-1} dp_1}{\left\{ \sum_{j=1}^{r_1} t_{1j}^{p_1} + (n-r-k)T^{p_1} + y^{p_1} \right\}^{r_1+1}} \right. \\ \int_0^\infty \frac{p_2^{r_2} \lambda_2^{p_2-1} dp_2}{\left\{ \sum_{j=1}^{r_2} t_{2j}^{p_2} + kT^{p_2} \right\}^{r_2} + r_2(r_2+k+1)} \\ \int_0^\infty \frac{p_1^{r_1} \lambda_1^{p_1-1} dp_1}{\left\{ \sum_{j=1}^{r_1} t_{1j}^{p_1} + (n-r-k)T^{p_1} \right\}^{r_1}} \\ \left. \int_0^\infty \frac{p_2^{r_2+1} \lambda_2^{p_2-1} y^{p_2-1} dp_2}{\left\{ \sum_{j=1}^{r_2} t_{2j}^{p_2} + kT^{p_2} + y^{p_2} \right\}^{r_2+1}} \right]$$

where

$$(3.4) \quad C^{-1} = (n+2) \sum_{k=0}^{n-r} \binom{n-r}{k} B(n-k-r_2+1, r_2+k+1) \\ \left[\int_0^\infty \frac{p_1^{r_1} \lambda_1^{p_1-1} dp_1}{\left\{ \sum_{j=1}^{r_1} t_{1j}^{p_1} + (n-r-k)T^{p_1} \right\}^{r_1}} \right. \\ \left. \int_0^\infty \frac{p_2^{r_2} \lambda_2^{p_2-1} dp_2}{\left\{ \sum_{j=1}^{r_2} t_{2j}^{p_2} + kT^{p_2} \right\}^{r_2}} \right].$$

$(1-\gamma)$ -prediction limits (P_1, P_2) are solutions of the equations

$$(3.5) \quad \int_0^{P_1} h(y|\underline{t}) dy = \int_{P_2}^\infty h(y|\underline{t}) dy = \gamma/2$$

where $h(y|\underline{t})$ and C are defined in (3.3) and (3.4).

4. NUMERICAL EXAMPLE

We demonstrate the preceding section's algebra with an example from Lawless (1982), page 202, dealing with polyethylene cable insulation. The data comprised 10 specimens each from two cable types subjected to accelerated life testing and it was reasonable to assume an underlying Weibull process.

Type 5.1, 9.2, 9.3, 11.8, 17.7, 19.4, 22.1, 26.7, 37.3, 60.0

I:

Type 11.0, 15.1, 18.3, 24.0, 29.1, 38.6, 44.2, 45.1, 50.9, 70.0

II:

Empirical goodness of fit tests (D'Agostino and Stephens, 1986) were used to assess the Weibull model assumption. Tests were performed to confirm that each sample was likely to have arisen from a Weibull process (p -values of 0.47 and 0.92 respectively), and that the overall Weibull mixture distribution was reasonable (p -value of 0.88). Parametric values were not available for the given samples so maximum likelihood estimates were calculated for use in constructing the prediction interval. At this point it was discovered that the parameterization used in this paper is less well

suitable to practical applications than the more commonly used form seen in Lawless (1982), for example. The difference lies merely in our definition of θ , which in the Lawless form is θ^p . For our example, the mle's under the more common parameterization are $p_1 = 1.51$, $p_2 = 2.11$, $\theta_1 = 24.40$, $\theta_2 = 39.25$, whereas, under our parameterization the θ 's become $\theta_1 = \theta_1^{p_1} = (24.40)^{1.51} = 124.44$ and $\theta_2 = \theta_2^{p_2} = (39.25)^{2.11} = 2306.76$. This discrepancy does not alter the algebra in the previous section, however, it does have implications for computer implementation.

The 95 % prediction interval was found to be (2.184, 62.500). The predictive density function was of a nature analogous to those portrayed in the next section, so we will not present it here. The final observation in each of the samples was then censored, resulting in a wider predictive density function (see next section), and an augmented prediction interval of (2.023, 73.456).

5. MONTE CARLO SIMULATION

Sample simulation was performed with the aid of the IMSL routines GGUBS and REXP for the generation of uniform and exponential variates respectively. The uniform data points were used to determine the sub-population membership and the exponential variates were then transformed into Weibull deviates following Marsaglia (1961). As in the previous section, empirical goodness of fit tests were used to check that the resulting samples could in fact have arisen from a mixture of Weibull models.

The algorithm was implemented on the AMDAHL 470 mainframe at the University of Manitoba, using the PL/I programming language. The algorithm for the censored sample was much more expensive than that of the complete sample counterpart, as might be expected.

We generated samples of different size from the mixture distribution (1.1) with different censoring point (or percentage of information used: PIU) and computed 95 % prediction intervals corresponding to (2.7) and (3.5). The results are given in Table 1.

TABLE 1
WIDTH OF PREDICTION INTERVALS GIVEN n AND PIU

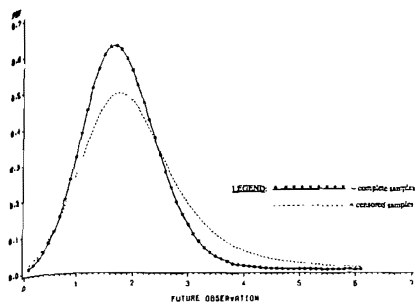
$n \backslash$ PIU	40	50	60	70	100
10	6.44	5.22	3.75	3.24	2.23
15	6.40	4.02	3.75	2.87	2.58
20	6.18	3.18	2.82	2.69	2.34
30	5.04	3.96	2.88	2.56	2.57
40	2.52	2.87	2.77	2.88	2.88
50	2.25	2.35	2.84	3.83	3.05

We observe in general that the effect of censoring is much more pronounced for small samples ($n < 30$). Specifically,

- (i) the prediction intervals for small censored samples ($n < 30$) are noticeably wider than those for the complete sample case which is what one would expect since we have only partial information with censored samples. As n increases, however, the two distributions converge.
- (ii) for small fixed n , as PIU increases the width of the prediction interval decreases and ultimately tends to the uncensored prediction interval;
- (iii) For larger n more information remaining after censoring produces more stable interval estimates.

The corresponding predictive distributions (2.5) and (3.3) are plotted in Figs. 1-6.

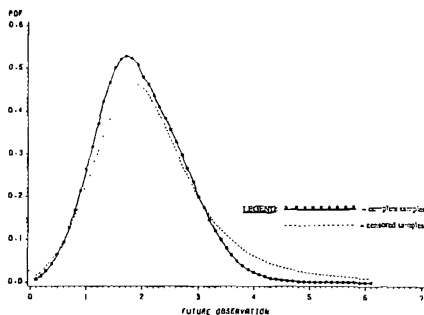
PREDICTIVE DENSITY FUNCTION
MIXTURE OF WEIBULL DISTRIBUTIONS
COMPLETE AND CENSORED SAMPLES CASE



PARAMETERS $P_1 = 2, P_2 = 3, \theta_1 = 3, \theta_2 = 6, n = 10$

Fig. 1

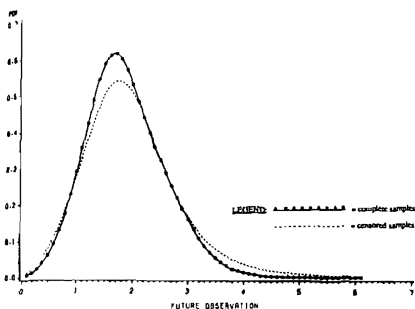
PREDICTIVE DENSITY FUNCTION
MIXTURE OF WEIBULL DISTRIBUTIONS
COMPLETE AND CENSORED SAMPLES CASE



PARAMETERS $P_1 = 2, P_2 = 3, \theta_1 = 3, \theta_2 = 6, n = 15$

Fig. 2

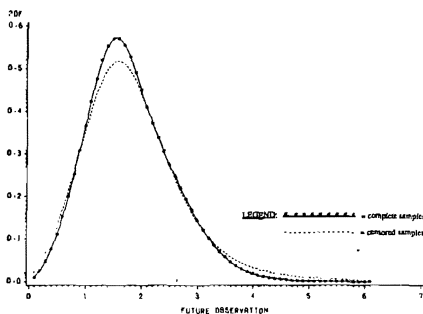
PREDICTIVE DENSITY FUNCTION
MIXTURE OF WEIBULL DISTRIBUTIONS
COMPLETE AND CENSORED SAMPLES CASE



PARAMETERS $P_1 = 2, P_2 = 3, \theta_1 = 3, \theta_2 = 6, n = 20$

Fig. 3

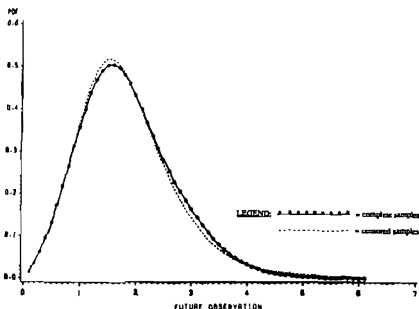
PREDICTIVE DENSITY FUNCTION
MIXTURE OF WEIBULL DISTRIBUTIONS
COMPLETE AND CENSORED SAMPLES CASE



PARAMETERS $P_1 = 2, P_2 = 3, \theta_1 = 3, \theta_2 = 6, n = 30$

Fig. 4

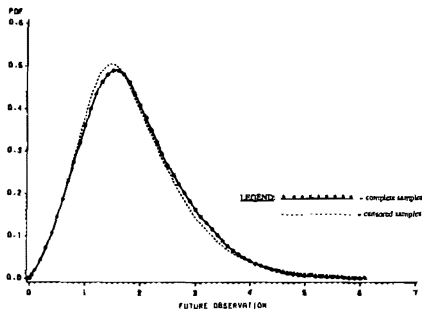
PREDICTIVE DENSITY FUNCTION
MIXTURE OF WEIBULL DISTRIBUTIONS
COMPLETE AND CENSORED SAMPLES CASE



PARAMETERS $P_1 = 2, P_2 = 3, \theta_1 = 3, \theta_2 = 6, n = 40$

Fig. 5

PREDICTIVE DENSITY FUNCTION
MIXTURE OF WEIBULL DISTRIBUTIONS
COMPLETE AND CENSORED SAMPLES CASE



PARAMETERS $P_1 = 2, P_2 = 3, \theta_1 = 3, \theta_2 = 6, n = 50$

Fig. 6

6. CONCLUSIONS

In this paper we investigated the use of Bayes techniques to predict future observations from a mixture of two Weibull distributions. Comparison of the predictive density functions for complete and censored samples showed that as the sample size increased, the loss of information due to censoring has a decreasing impact on the shape of the distribution. The subsequent Bayes predictive intervals were considerably wider in the censored case for small samples. As n increased, however, the amount of information retained after censoring was enough to produce predictive intervals with a comparable width to those in the complete samples case. This holds practical implications for the use of censoring for large samples in order to predict future observations.

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